

Solution Requirements

Date	29 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	The system does not require user authentication. Users can directly access the application to predict the Human Development Index (HDI).	Low
2	Authorization Levels	No user roles are implemented. All users have equal access to use the prediction system.	Low
3	External Interfaces	The application uses the trained Machine Learning model, Flask framework, and HTML/CSS interface. No external APIs are required.	High
4	Transaction Processing	The system accepts user input, processes it using the trained Linear Regression model, and returns the predicted HDI value with the corresponding Human Development category.	High

S.No	Requirement Category	Requirement Description	Priority
5	Reporting	The application displays the predicted HDI value and Human Development category on the results page.	Medium
6	Business Rules	User inputs must be valid. The trained model predicts HDI using selected development indicators and classifies the result into Low, Medium, High, or Very High Human Development.	High
7	Compliance to Laws or Regulations	The application does not collect or store personal information and follows general data privacy principles for publicly available datasets.	Medium
8	Other	The application provides a simple and user-friendly interface for instant HDI prediction.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should generate HDI predictions quickly after receiving user input.	Prediction displayed within 2 seconds .
2	Scalability	The application should support multiple users accessing the prediction service simultaneously.	Supports multiple concurrent users without noticeable delay.
3	Security & Data Privacy	The application should securely process user input without storing personal data.	No sensitive user data stored; secure request handling.
4	Reliability & Availability	The prediction system should be available whenever users access the application.	99% uptime during deployment.
5	Usability & Accessibility	The web application should provide an intuitive and responsive interface for users.	Easy navigation with clear input fields and prediction results.

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
6	Other (Maintainability)	The code should be modular and easy to update with new datasets or improved models.	Model and application can be updated without major code changes.